Compact VEP-EEG Recognition: From Classical/Deep Learning Baselines to Spiking Neural Network Enhancement

A working project brief for a joint master's-thesis collaboration

Nhan Tran* Jenni Raitoharju†

October 9, 2026

Abstract

Compact, low-channel wearable EEG (e.g., 14-channel Emotiv EPOC-X) is increasingly used for visual stimulus recognition, but most reported gains come from either heavier classical feature engineering or from deep architectures whose benefit over simpler baselines is rarely isolated cleanly. This project narrows the scope to a single, sharply framed question: *under a controlled, fair benchmark, how much recognition performance can be gained by adding spiking-neural-network (SNN) temporal dynamics on top of standard classical machine-learning (ML) and deep-learning (DL) baselines for 4-class visual stimulus recognition from compact VEP-EEG?* We fix data, folds, preprocessing, and evaluation protocol across all models, so that any improvement from the SNN stage can be attributed to the temporal-dynamics mechanism itself rather than to confounds in preprocessing or evaluation. This document describes the idea, what has already been done, what remains, the data source, and what we would expect from a master's-thesis collaborator joining the project.

1 Idea and Motivation

Visually evoked potentials (VEPs) recorded with compact, consumer-grade EEG devices are short, noisy, and low-channel-count compared to research-grade EEG. Two current practices are common in the literature:

(i) Reporting a deep model's accuracy on a VEP task without a matched classical-ML baseline, making it unclear whether the deep model actually adds value beyond simple features and a linear or kernel classifier.

(ii) Reporting a spiking-neural-network result without isolating *which part* of the pipeline (feature extraction vs. temporal spiking dynamics vs. the classification head) is responsible for the reported gain.

*Doctoral Researcher, Faculty of Information Technology, University of Jyväskylä (JYU), Finland. Corresponding author: **[TODO: email]**

†Associate Professor, Machine Intelligence Methods and Applications (MIMA) group, University of Jyväskylä (JYU), Finland.

This project addresses both gaps directly. We restrict the scope to two families of models, evaluated on the exact same folds and preprocessing:

- **Classical ML and conventional DL baselines** — feature-based classifiers (e.g., LinearSVC, RBF-SVC, PCA+RBF-SVC) and standard EEG deep networks (e.g., EEGNet-style and ShallowConvNet-style convolutional architectures), trained and evaluated end to end on the 4-class task.
- **SNN enhancement** — spiking (leaky integrate-and-fire, LIF) temporal dynamics applied on top of learned convolutional feature maps, tested both as (a) a frozen spike-count representation followed by a classical probe, and (b) a supervised, end-to-end SNN-hybrid model with a differentiable prototype/RBF-style classification head.

The central claim we want to establish (or refute, honestly) is: *adding LIF temporal dynamics on top of a simple convolutional feature extractor (ShallowConvNet) is a clearer route to improved compact-VEP recognition than making the convolutional backbone itself larger or more complex.* Everything else — quantum-classical models, foundation-model transfer, interface-alignment taxonomy — is intentionally out of scope for this narrower project so that the ML/DL-vs-SNN comparison stays clean and easy to defend.

2 Contributions

C1. A controlled ML/DL baseline benchmark for compact VEP-EEG. We build a unified benchmark for 4-class visual stimulus recognition from 14-channel, short-window VEP-EEG. It spans classical ML baselines (LinearSVC, RBF-SVC, PCA+RBF-SVC on handcrafted and learned features) and standard deep EEG architectures, all evaluated under one fixed five-fold subject-aware protocol. Every preprocessing step that could leak information across folds (scaling, dimensionality reduction, probe fitting) is restricted strictly to the training fold in each split.

C2. SNN temporal-dynamics enhancement, isolated and measured on top of the same baselines. On top of the frozen feature maps learned by the best conventional baseline (ShallowConvNet), we apply LIF temporal dynamics and test two ways of using them:

- *Representation pathway*: spike-count readout from a temporal LIF module, followed by a nonlinear probe (RBF-SVC) — no additional supervised training of the SNN stage itself.
- *Supervised pathway*: an end-to-end SNN-hybrid model that fuses spike-count features with a differentiable prototype/RBF-style classification head, trained jointly.

We report both pathways separately so the "free" representation-level gain (no extra training) is never conflated with the gain from further supervised training of the SNN stage.

C3. Cross-dataset generalization of the same ML/DL-vs-SNN pipeline. We apply the identical pipeline from C1–C2 (fixed folds, train-only preprocessing, ML baselines, DL baseline, frozen SNN representation probe, supervised SNN-hybrid) to a second, independent EEG classification task, to test whether the SNN gain observed on 4-class VEP recognition also holds on a different signal type and a different classification target, rather than being specific to visual-stimulus VEP data.

3 Data

Dataset. We use the public dataset *"EEG Dataset for Natural Image Recognition through Visual Stimuli"*, hosted on Mendeley Data (Birla Institute of Technology, funded by India's Science and Engineering Research Board).

- **Link / DOI:** https://data.mendeley.com/datasets/g9shp2gxhy/2 (DOI: `10.17632/g9shp2gxhy.2`, published 20 January 2025, Version 2).
- **Device:** 14-channel Emotiv EPOC-X, 10–20 electrode system, sampled at 128 Hz, recorded and annotated with EmotivPRO software.
- **Subjects:** 35 adult participants screened with the Vividness of Visual Imagery Questionnaire (VVIQ); 32 participants cleared the screening and have usable EEG.
- **Task / classes (4-class):** participants viewed natural images from four classes while their VEP-EEG was recorded — **A**pple, **F**lower, **C**ar, and human **P**face(face). Each class folder contains per-participant subfolders with both `.csv` and `.edf` recordings.
- **Metadata:** `Participant_info.csv` (age, gender, image class viewed, VVIQ score, anonymized participant ID) and `VVIQuestionnaire.pdf` (the questionnaire instrument itself).

Access. The dataset is open-access on Mendeley Data; no application or data-use agreement is required beyond standard attribution.

Preprocessing plan. 1-second VEP windows are extracted per trial; all scaling and dimensionality-reduction fitting happens on the training fold only, inside a fixed five-fold (subject-aware) cross-validation protocol, to avoid any leakage into the held-out fold.

Two candidate datasets for cross-dataset generalization (C3), similar in spirit to the primary dataset above. Rather than switching task families, we look for datasets that keep the same recipe — compact, consumer-grade EEG, a discrete visual-stimulus classification target — so the C1–C2 pipeline transfers with minimal changes.

- **Candidate 1: Emotiv EPOC+ VEP/SSVEP/motor-imagery dataset** (Fernandez-Fraga, Aceves-Fernandez & Pedraza-Ortega, *Data in Brief*, 2019, DOI: `10.1016/j.dib.2019.103871`; hosted on the UCI Machine Learning Repository as *"EEG Steady-State Visual Evoked Potential Signals"*). 29 subjects, 14-channel Emotiv EPOC+ (same electrode family as our primary device), 10–20 system. Includes a VEP visual-search task and an SSVEP "five-box" task with a clear attended-vs-unattended discrete label per trial — directly usable as a classification target with the same preprocessing and fold logic as C1–C2.
- **Candidate 2: MindBigData "Visual MNIST of Brain Digits"** (https://www.mindbigdata.com/opendb/). EEG recorded with, among other devices, the 14-channel Emotiv EPOC while subjects viewed real MNIST digit images (10 discrete classes, digits 0–9), i.e. the same discrete-visual-stimulus-classification shape as our primary 4-class task, just with more classes and a different stimulus type (digits vs. natural images).

Caveat to resolve before committing to either candidate. Our primary dataset's protocol uses subject-aware five-fold cross-validation (Section 3), which needs enough independent subjects per fold. Candidate 1 has 29 subjects and fits directly. Candidate 2's MindBigData recordings are collected mostly from a single test subject, so the same subject-aware split cannot be reused as-is; if we pick Candidate 2, the fold protocol needs a documented adaptation (e.g. trial-level stratified folds instead of subject-level) —

4 What Has Been Done So Far

- Data pipeline: dataset downloaded, 1-second VEP windows extracted, fixed five-fold subject-aware split defined and frozen.
- Classical ML baselines: LinearSVC, RBF-SVC, and PCA+RBF-SVC implemented and run to completion on the 4-class task.
- Conventional DL baseline: ShallowConvNet-style architecture trained and validated under the same fold protocol; identified as the strongest conventional baseline so far.
- SNN representation pathway: LIF temporal dynamics applied to ShallowConvNet's learned feature maps; spike-count readout extracted and probed with RBF-SVC, showing promising gains over the frozen-feature baseline without any extra supervised training.
- Initial version of the supervised SNN-hybrid model (spike-count fusion + prototype/RBF-style head) implemented; early runs completed but not yet fully tuned or statistically validated.

5 Results So Far: ML, DL, and SNN Only

Table 1 reports the five-fold C1 benchmark results already obtained on the primary VEP-EEG dataset. *Quantum–classical hybrids and self-supervised EEG foundation-model transfer results exist from an earlier, broader draft of this project but are intentionally left out here, since C1–C3 as scoped in this brief only cover classical ML baselines, the conventional DL baseline, and SNN enhancement.*

The pattern that matters for the central claim in Section 1: every SNN row already beats the best conventional DL baseline (ShallowConvNet, 82.14% accuracy), and the best SNN representation pathway (LIF spike-count + RBF, 86.02%) does so *without any supervised fine-tuning of the SNN stage itself* — it is a frozen probe on top of ShallowConvNet's own features. This is the empirical basis for C2, and it is exactly the comparison we plan to repeat on the two candidate datasets in Section 3 for C3.

6 Next Steps

1. **Finish tuning the supervised SNN-hybrid model** — hyperparameter search over LIF time constants, spike threshold, number of timesteps, and prototype/RBF head parameters, under the same five-fold protocol used for every other model.
2. **Statistical comparison** — paired significance testing (e.g., per-fold paired t-test or Wilcoxon signed-rank) between (a) ML baselines vs. DL baseline, and (b) DL baseline vs. both SNN pathways, to make sure reported gains are not fold-to-fold noise.

Table 1: Five-fold C1 benchmark results on compact 14-channel VEP-EEG, restricted to classical ML baselines, the conventional DL baseline, and SNN representations/hybrids (quantum-classical and foundation-model rows omitted). Values are mean ± sample standard deviation. Bold marks the best result within each family.

Model	Acc.	Macro-F1	Bal. Acc.	$\kappa \times 100$
Flattened-feature classical baselines				
Random Forest	**44.57±2.03**	**41.29±2.02**	**41.60±1.94**	**23.57±2.89**
LightGBM	44.16±2.11	41.98±2.21	41.86±2.04	23.48±2.82
XGBoost	43.80±2.56	40.89±2.59	41.06±2.47	22.74±3.41
SVM (RBF, raw)	41.86±0.97	37.41±1.71	38.51±1.24	19.50±1.58
AdaBoost	34.03±1.79	30.02±1.57	31.09±1.63	8.63±2.49
Decision Tree	30.12±1.66	29.33±1.58	29.37±1.57	5.71±2.18
Logistic Regression	26.41±1.13	25.82±1.03	25.85±1.05	0.87±1.55
Conventional EEG neural baselines				
DeepConvNet	77.13±2.07	75.83±2.11	76.28±2.53	69.11±2.85
ShallowConvNet	**82.14±1.01**	**81.33±0.88**	**81.20±0.81**	**75.88±1.34**
Spiking neural representations (C2)				
Temporal LIF head	82.56±1.57	81.82±1.44	81.62±1.41	76.44±2.09
Temporal gated LIF	83.50±0.89	82.83±0.84	82.78±0.80	77.74±1.18
Spike-aware gated LIF	84.41±1.29	83.73±1.13	83.67±1.13	78.96±1.73
Prototype LIF	84.75±0.66	83.94±0.51	83.85±0.41	79.40±0.85
LIF spike-count + PCA/RBF	85.71±0.86	85.04±0.70	84.88±0.70	80.71±1.16
LIF spike-count + RBF	**86.02±0.93**	**85.39±0.82**	**85.29±0.87**	**81.12±1.25**

3. **Ablations** — remove/vary one SNN component at a time (spike-count readout only, prototype head only, number of LIF timesteps) to attribute the gain to a specific mechanism rather than to the SNN stage as a whole.

4. **Robustness checks** — confirm results hold under a second, independent random seed set and under a leave-one-subject-out variant of the protocol, to rule out overfitting to the specific five-fold split.

5. **Compute/efficiency comparison** — report inference latency and parameter count for the SNN-hybrid model against the DL baseline, since a practical motivation for SNNs is efficiency, not only accuracy.

6. **Writing** — full first draft (Introduction, Related Work, Method, Experiments, Results, Discussion, Conclusion), figures for the pipeline and for the fold-wise result comparison, and a clean, reproducible code release.

7. **Target venue** — **[TODO: Nhan and Jenni to decide: workshop paper, EEG/BCI-focused journal, or a general ML venue with an EEG application track.]**

8. **Cross-dataset generalization (C3)** — apply the exact same method already used and validated on the primary dataset (Table 1: classical ML baselines, the ShallowConvNet DL baseline, then both SNN pathways) to *both* candidate datasets in Section 3, under an analogous fixed-fold protocol, each on its own discrete visual-stimulus classification target.

7 What We Would Expect from a Master's-Thesis Collaborator

This project is scoped to be a realistic, well-defined master's thesis: the data is ready, the baselines exist, and the open question (does SNN temporal dynamics reliably help, and why) is concrete and testable. We would expect the student to:

- Take ownership of **Section 6, items 1–5** (hyperparameter tuning, statistical testing, ablations, and robustness checks) as the core of their thesis work, under Jenni's supervision and in regular contact with Nhan.
- Work inside the existing, fixed five-fold evaluation protocol rather than redesigning it — the protocol is intentionally frozen so results stay comparable across the whole project.
- Keep code reproducible and documented (clear environment file, seeds fixed and reported, scripts to regenerate every table/figure from raw predictions).
- Contribute to the Method and Experiments sections of the eventual write-up, based on the experiments they personally ran.
- Bring their own initiative on the ablation design (Section 6, item 3) — this is the part of the project with the most room for independent thinking, and a good place for the student to make the work genuinely their own.
- Meet biweekly (every two weeks) to review progress against the next-steps list above.

In return, the student gets a scoped, already-validated project with working baselines, a clear open research question, real EEG data with no access barriers, and co-authorship on the resulting paper (author order and exact contribution split to be agreed once the ablation and tuning work is complete).